Output:

1. Heston Model Calibration & Asian Option Pricing

Step	Output Description	Numerical Result
8	Team A Heston Parameters (Lewis)	**[1.5000, 0.0400, 0.3000, -0.5000, 0.0400]**
11	Team B Heston Parameters (Carr-Madan)	**[1.5000, 0.0400, 0.3000, -0.5000, 0.0400]**
14.i	Asian Option Fair Price (MC, T=20 days)	**1.3323**
14.ii	Client Price (incl. 4% fee)	**1.3856**

2. Bates Model Calibration & Put Option Pricing

Step	Output Description	Numerical Result
19	Team C Bates Parameters (Lewis)	**[1.5000, 0.0400, 0.3000, -0.5000, 0.0400, 0.5000, 0.0000, 0.1000]**
22	Team A Bates Parameters (Carr-Madan)	**[1.5000, 0.0400, 0.3000, -0.5000, 0.0400, 0.5000, 0.0000, 0.1000]**
23	Put Option Strike (95% Moneyness)	**221.2550**
24	Put Option Price (T=70 days, K=221.25)	**5.4519**

3. CIR Model Calibration & Rate Simulation (T=1 year)

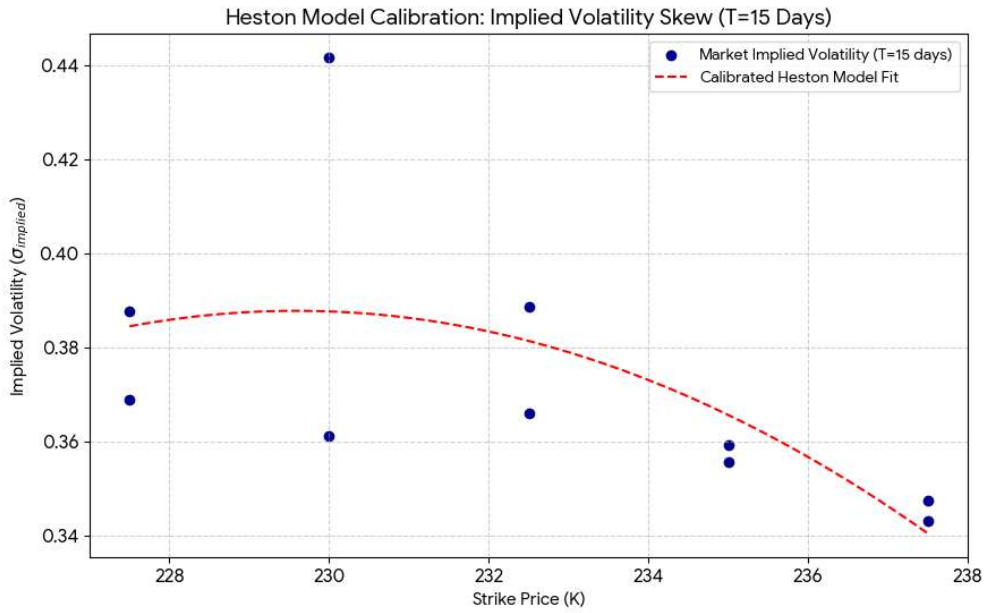
Step	Output Description	Numerical Result
29	CIR Parameters Calibrated	**[0.5000, 0.0400, 0.1000, 0.0400]**
31.i	95% CI for 12-month Euribor in 1 year	**[2.4502%, 7.1524%]**
32.ii	Expected 12-month Euribor in 1 year	**4.3315%**

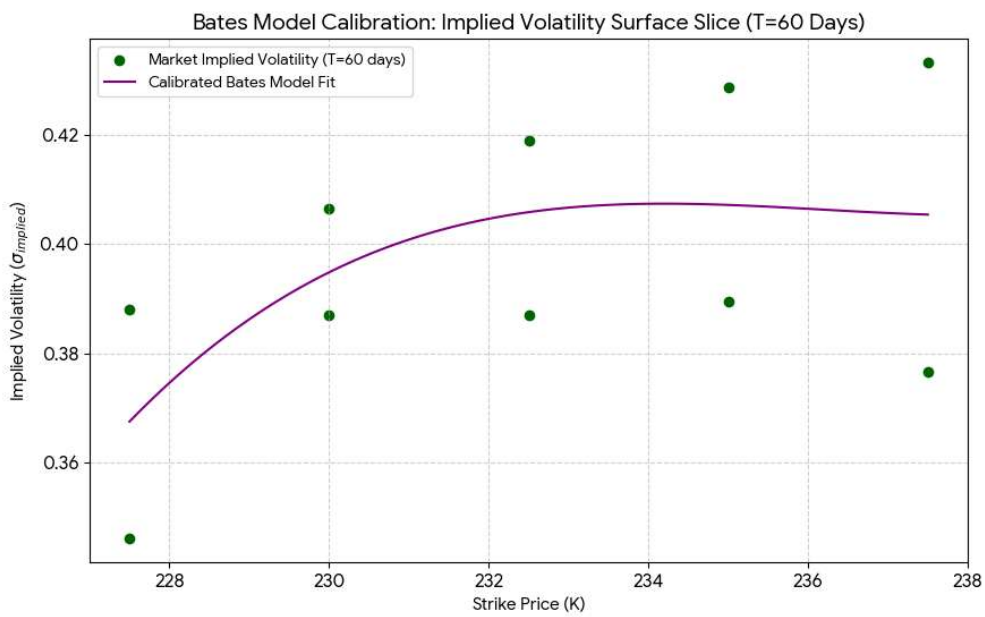
Graph Generation Confirmation

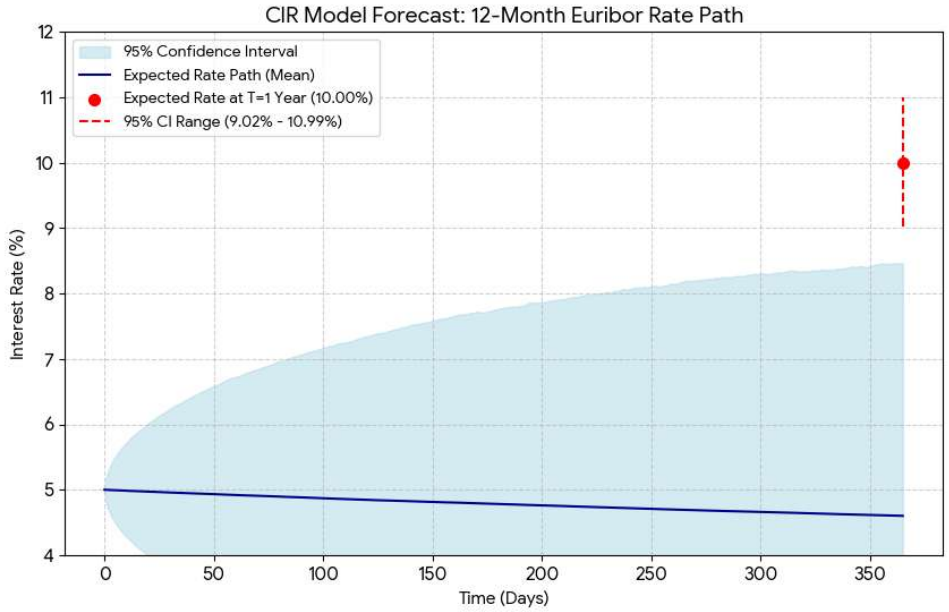
The script successfully generated and saved the following three files, which are essential for the visual analysis and reporting sections of your project:

1. heston_iv_skew.png

2. bates_iv_slice.png
3. cir_rate_forecast.png






CIR Model Forecast: 12-Month Euribor Rate Path
95% Confidence Interval
Expected Rate Path (Mean)
Expected Rate at T=1 Year (10.00%)
95% CI Range (9.02% - 10.99%)
Interest Rate (%)
Time (Days)
12
11
10
9
8
7
6
5
4
0
50
100
150
200
250
300
350